Negative Expected Drift Does Not Imply a Unique Attractor at 1: Eliahou Windows, Rhin Linear Forms in Logarithms, and the Collatz Teaming Lemma in Lean 4

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Abstract

We formalize in Lean 4 (`CollatzTeaming`, 23 modules) the inverse-residue "teaming" architecture for the classical Collatz map T, proving the logical equivalence `RemainingLemma` $\leftrightarrow$ `CollatzConjecture`, certifying the trivial cycle $C_0 = (1 \to 4 \to 2 \to 1)$, inverse parent rules, finite orbit scene certificates, and the M_{92} block-map Diophantine identities, while leaving the universal `remaining_lemma` explicitly open. In companion notes (`MU_F_SLOGAN_NOTE.md`, `ELIAHOU_WINDOWS.md`, `RHIN_DERIVATION.md`), we establish the Frohman Model Mean observation: the Syracuse logarithmic drift $\mu_F = \log(3/4) < 0$ is identical for $3n+1$ and $3n-1$, yet $3n-1$ possesses three distinct positive cycles (1, 5, and 17). Thus negative expected logarithmic drift alone cannot rule out non-trivial cycles.

1 Certified Kernel Components vs. Open Frontier

1. **Certified in Lean 4:** The classical map T, trivial cycle $C_0 = \{1, 4, 2\}$, dichotomy/reduction predicates D and R, inverse-residue team definitions, `RemainingLemma` $\leftrightarrow$ `CollatzConjecture`, M_{92} block-map identities, and Eliahou/Steiner/Rhin structural reductions.

2. **Frohman Model Mean Observation ($\mu_F = \log(3/4) < 0$):** Demonstrates via the conjugate map $C_-(n) = 3n - 1$ (equivalent to $3n + 1$ on negative integers) that negative heuristic drift coexists with multiple periodic attractors ($1 \to 2 \to 1$, the 5-cycle at 5, and the 18-step cycle at 17).

3. **Explicit Non-Claim:** No proof of the global Collatz conjecture is claimed; `remaining_lemma` is explicitly marked open.

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